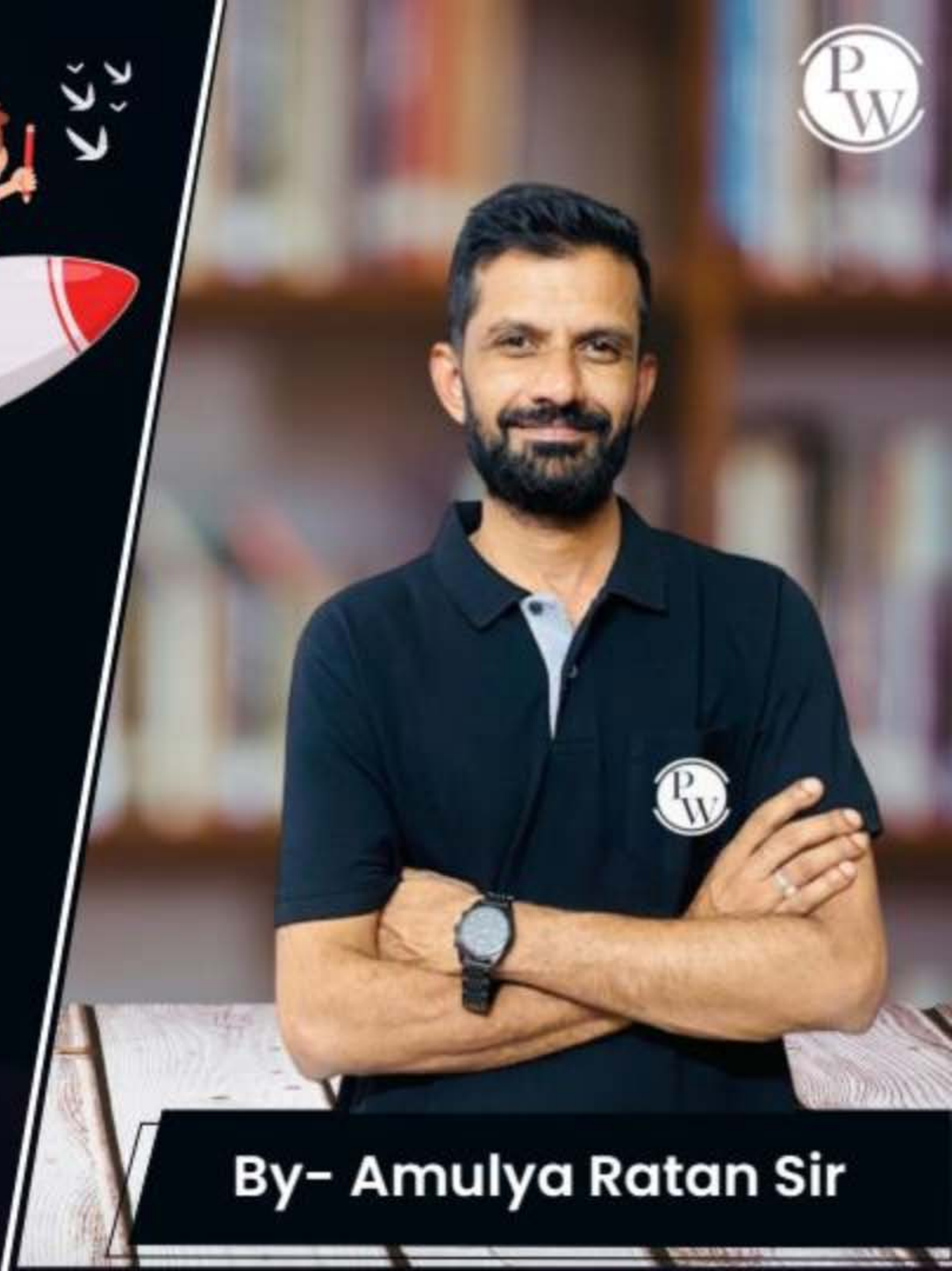
GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 16



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Paper Folding & Cutting

Topics to be Covered



Topic-1 Mensuration 2D & 3D

[MCQ]

#Q. Choose the alternative which is closely resembles the mirror image of the given combination.

MALAYALAM

Assignment

A. MALAYALAM

B. MALAYALAM

C. MALAYALAM

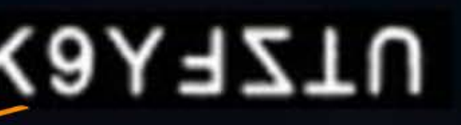
D. MALAYALAM

[MCQ]

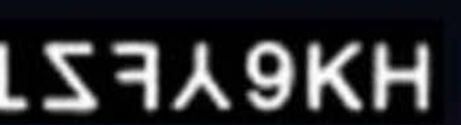
#Q. Choose the alternative which is closely resembles the mirror image of the given combination.

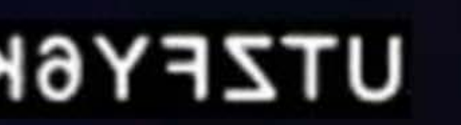
U T Z F Y 6 K H

Assignment

A 

B 

C 

D 

[MCQ]

#Q. Choose the alternative which is closely resembles the mirror image of the given combination.

SUPERVISOR

Assignment

A ꓤOSIVꓤƎꓭUS

B ꓤSOUNƧEꓤIV

C ƧUNEꓤVIꓴOꓤ

D ƧUNEꓤVIOSꓤ

[MCQ]

#Q. Choose the alternative which is closely resembles the mirror image of the given combination.

B R 4 A Q 1 6 H I

Assignment

A 

B 

C 

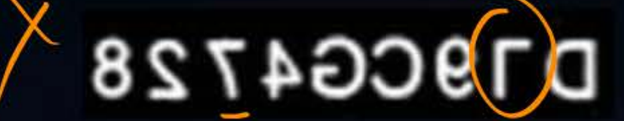
D 

[MCQ]

#Q. Choose the alternative which is closely resembles the mirror image of the given combination.

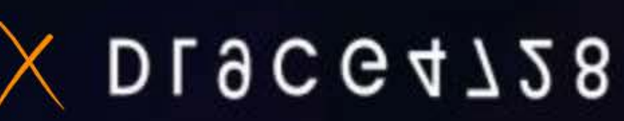
Assignment

D L 9 C G 4 7 2 8

A 

B 

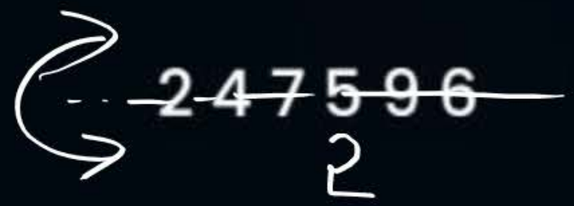
C 

D 

[MCQ]

#Q. Choose the alternative which is closely resembles the mirror image of the given combination.





Assignment

A ✓ (water Image)

B ✗

C ✗

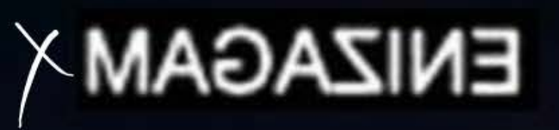
D ✗

[MCQ]

#Q. Choose the alternative which is closely resembles the mirror image of the given combination.

MAGAZINE

Assignment

A 

B 

C 

D 

[MCQ]

#Q. Choose the alternative which is closely resembles the mirror image of the given combination.





[MCQ]

#Q. Choose the alternative which is closely resembles the mirror image of the given combination.



A 

B 

C 

D 

Assignment

[MCQ]

#Q. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.











[MCQ]

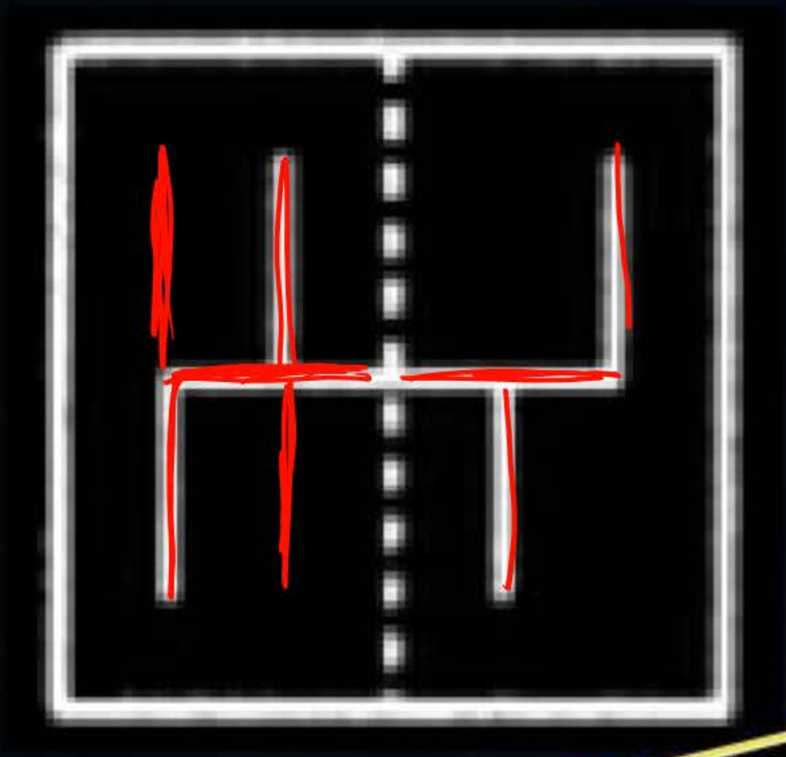
#Q. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.

A 

B 

C 

D 



Assignment

[MCQ]

#Q. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.



Assignment

[MCQ]

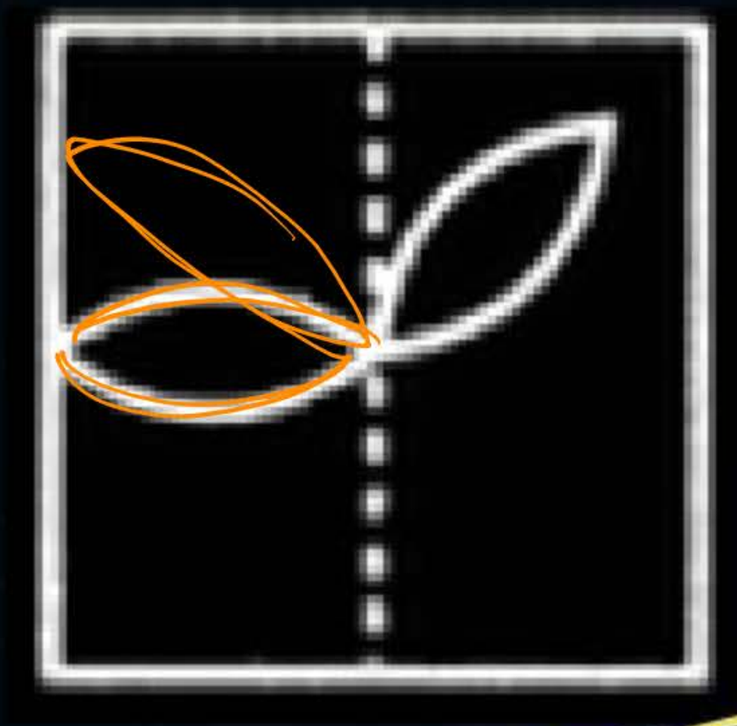
#Q. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.









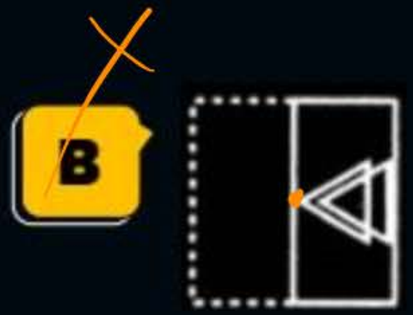


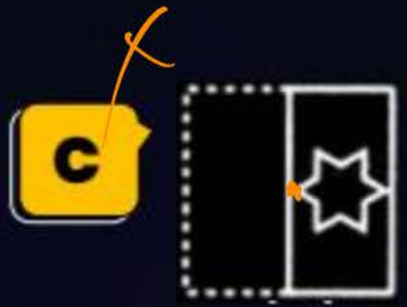
Assignment

[MCQ]

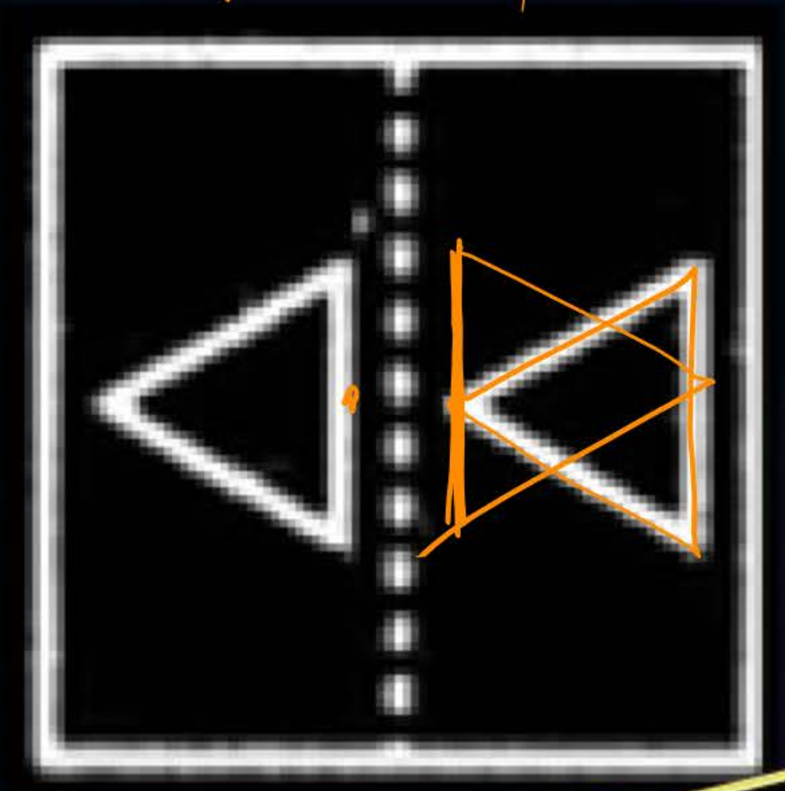
#Q. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.







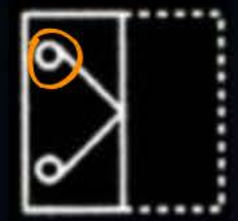


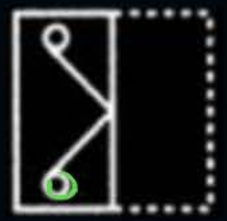


Assignment

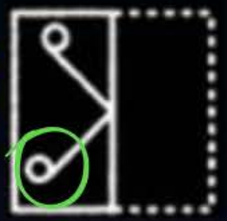
[MCQ]

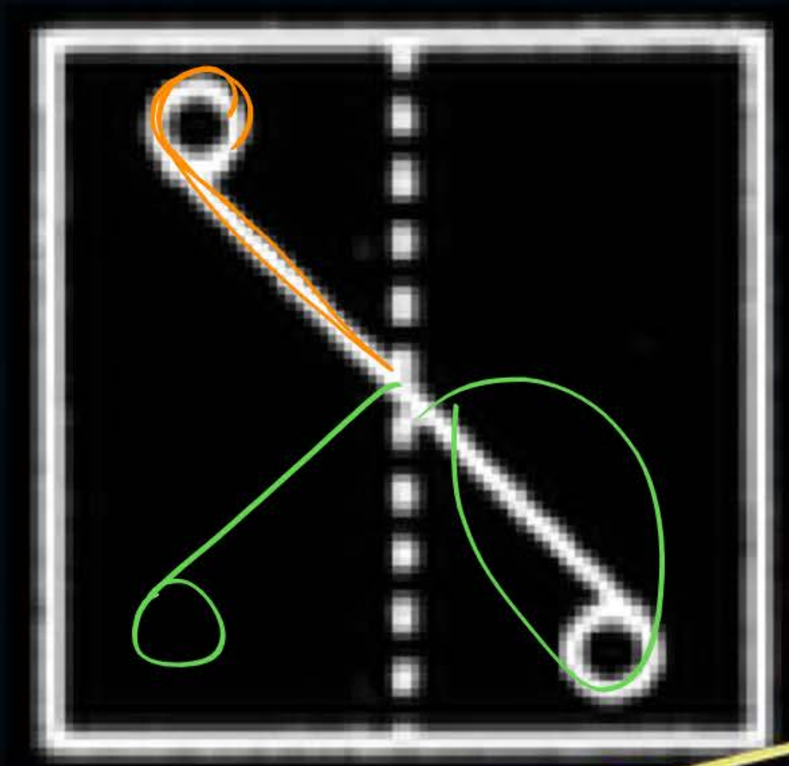
#Q. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.

A 

B 

C 

D 

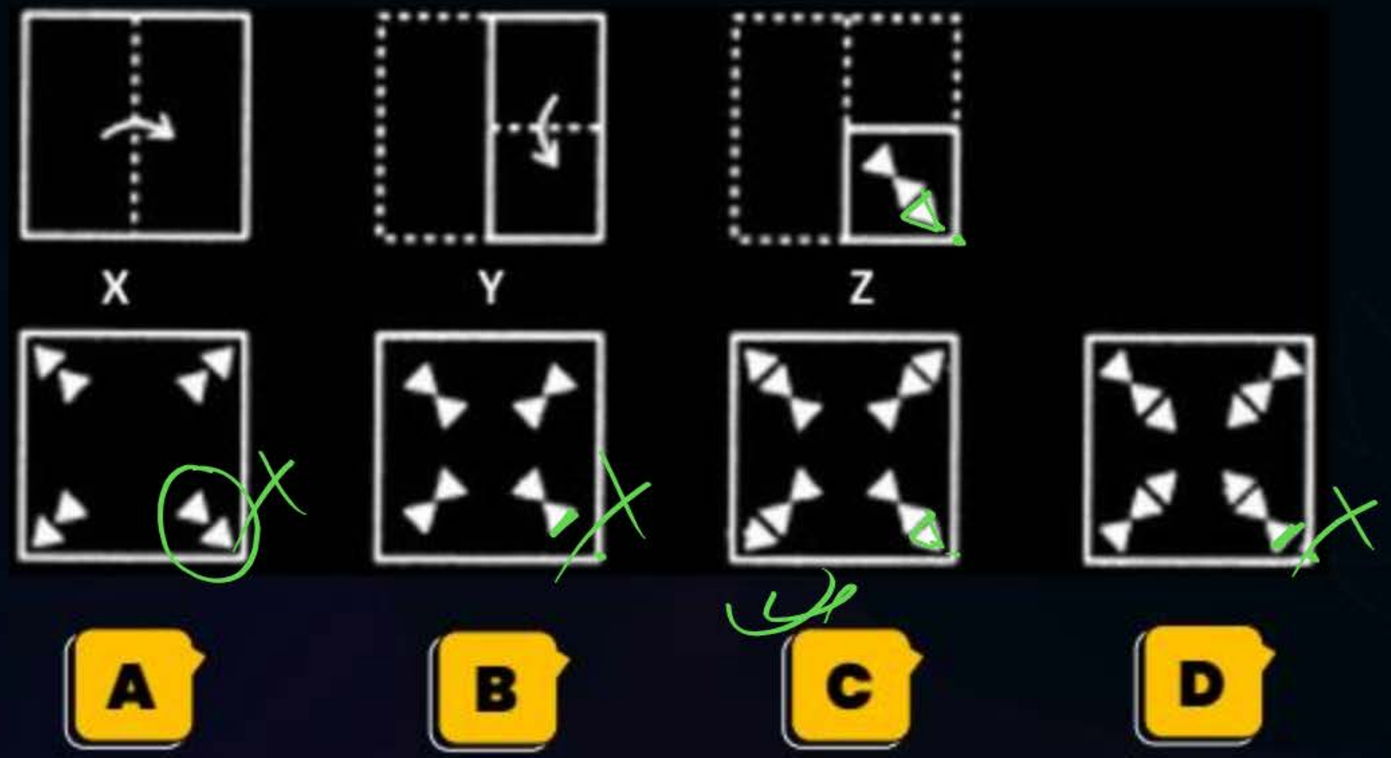


Assignment

[MCQ]

#Q. Choose a figure which would most closely resemble the unfolded form of Figure (Z).

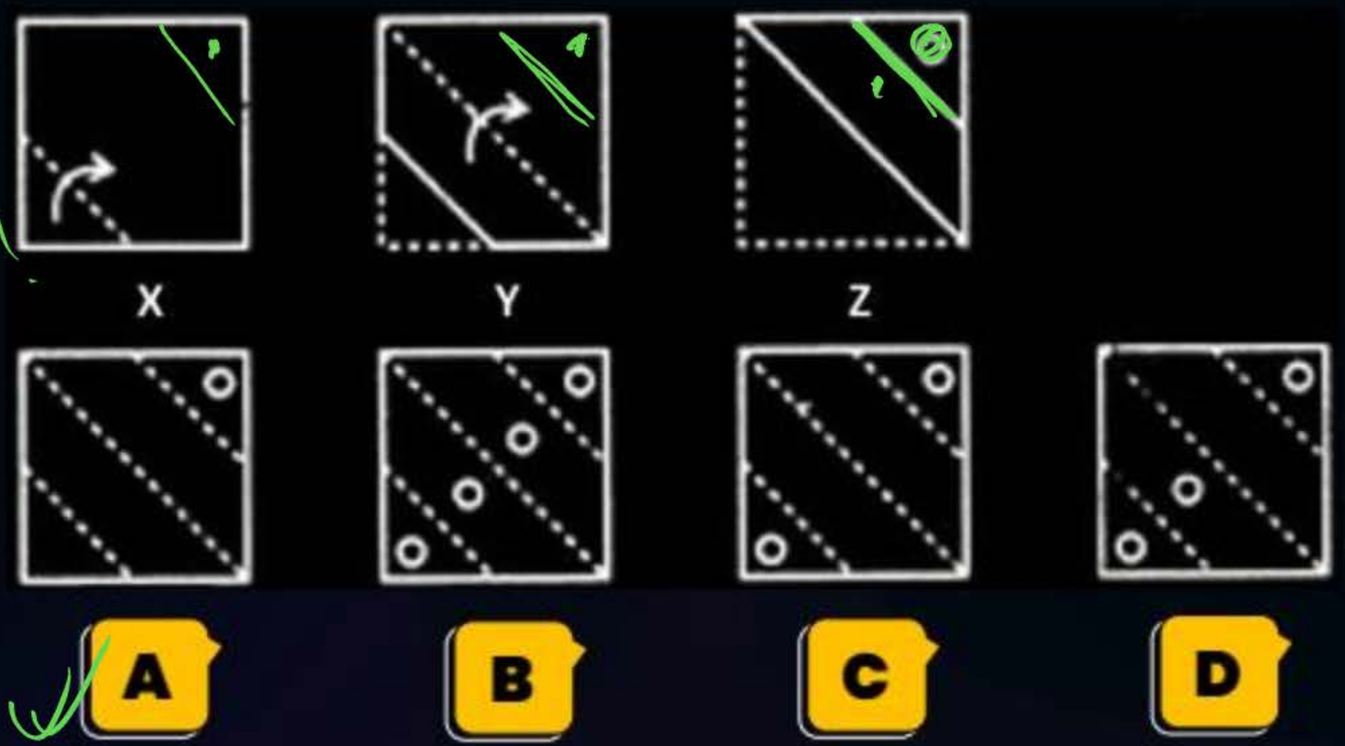
Assignment



[MCQ]

#Q. Choose a figure which would most closely resemble the unfolded form of Figure (Z).

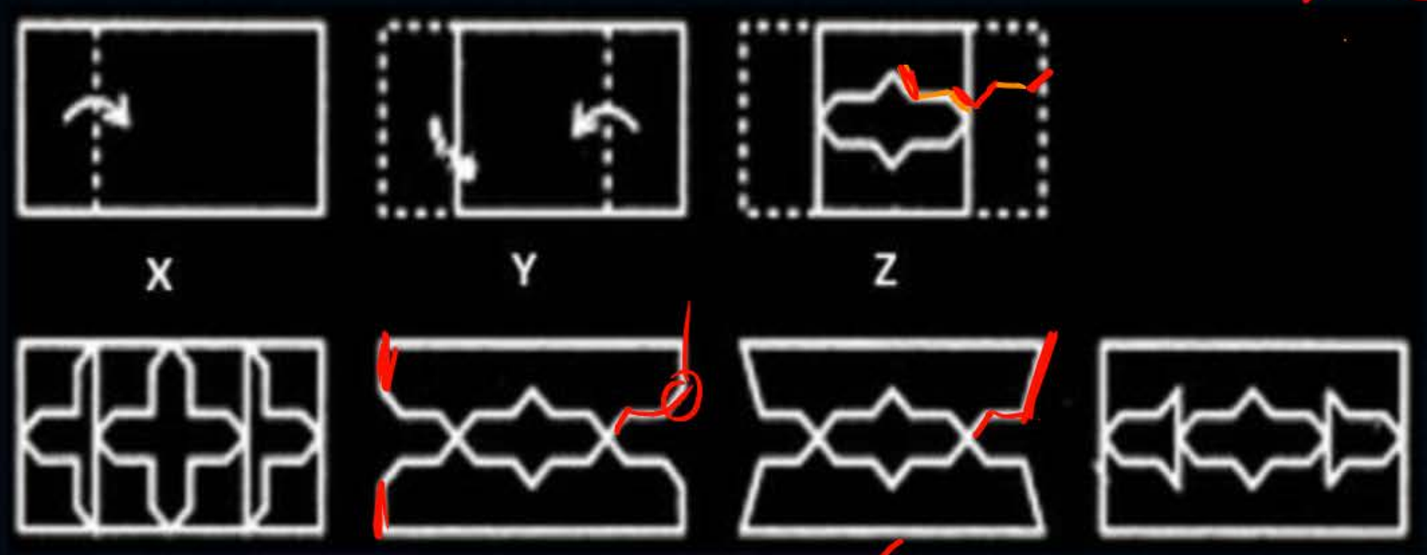
Assignment



[MCQ]

#Q. Choose a figure which would most closely resemble the unfolded form of Figure (Z).

Assignment

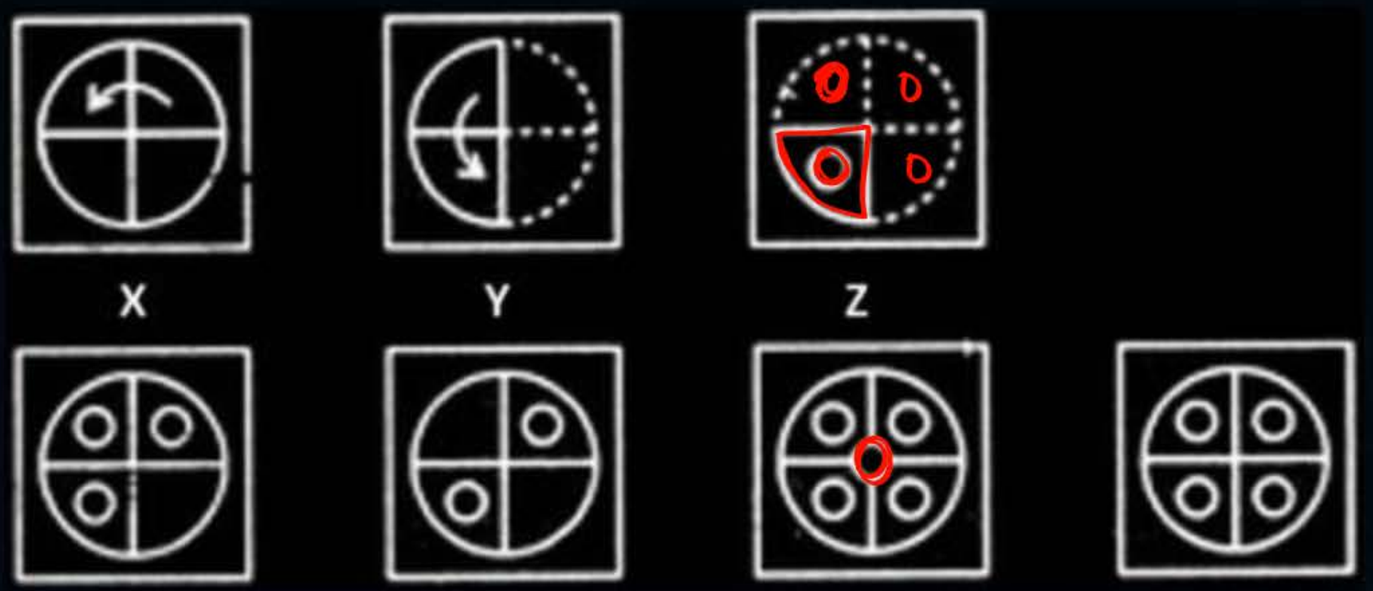


A B C D

[MCQ]

#Q. Choose a figure which would most closely resemble the unfolded form of Figure (Z).

Assignment

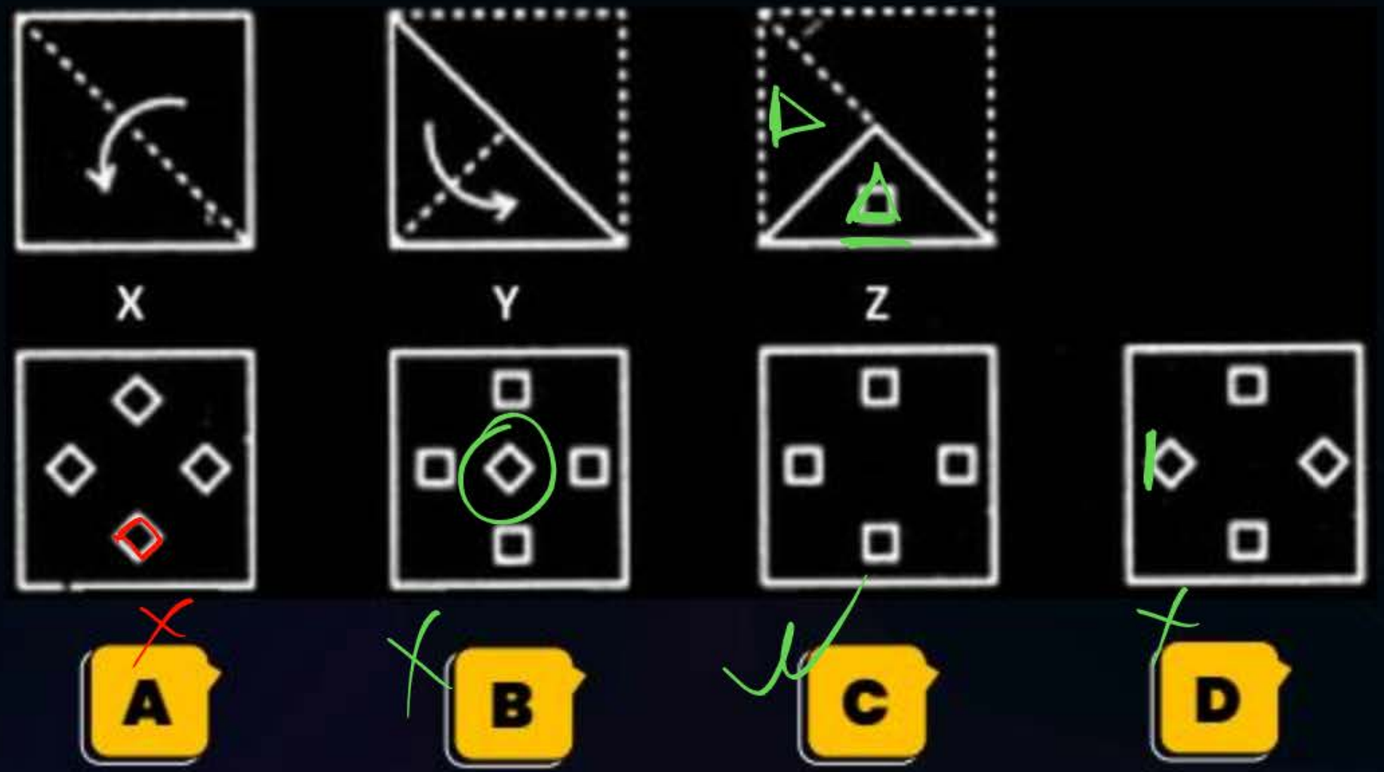


A B C D

[MCQ]

#Q. Choose a figure which would most closely resemble the unfolded form of Figure (Z).

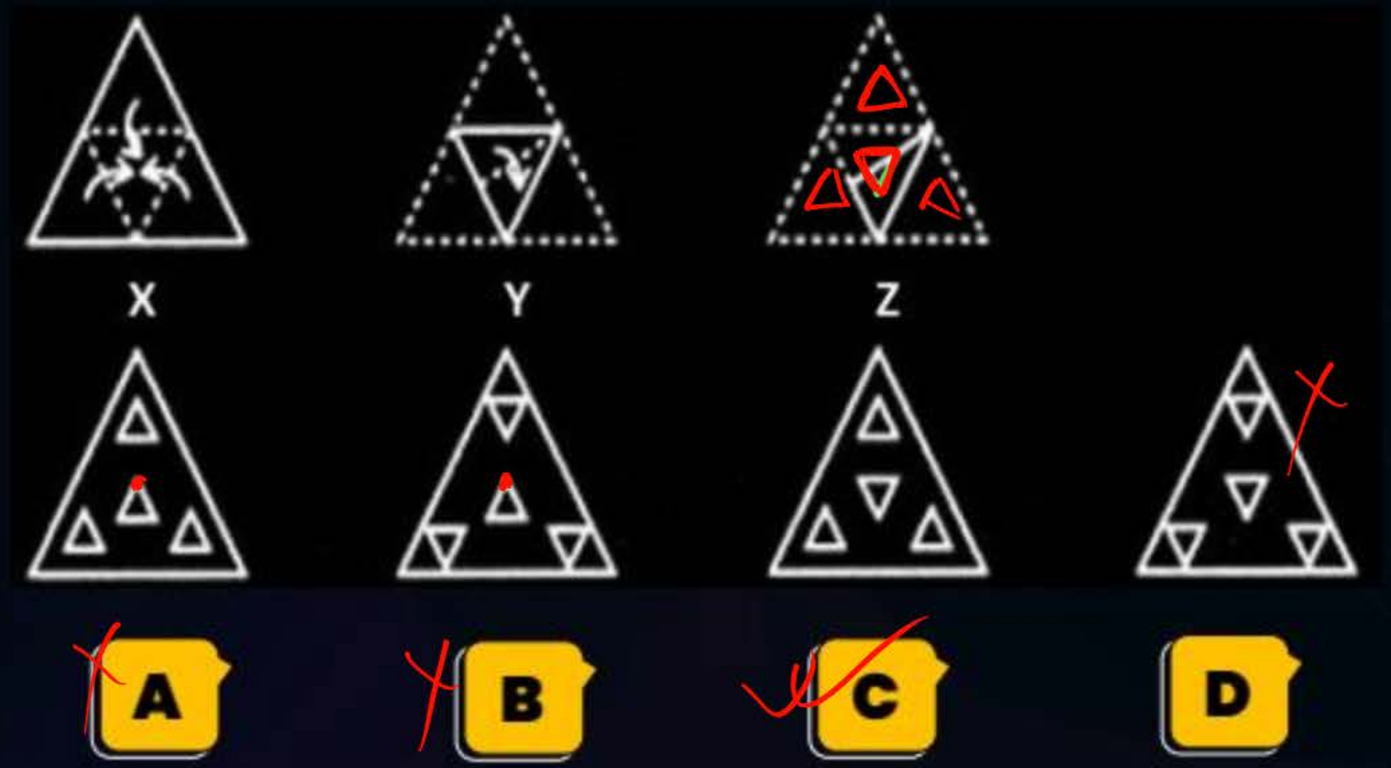
Assignment



[MCQ]

#Q. Choose a figure which would most closely resemble the unfolded form of Figure (Z).

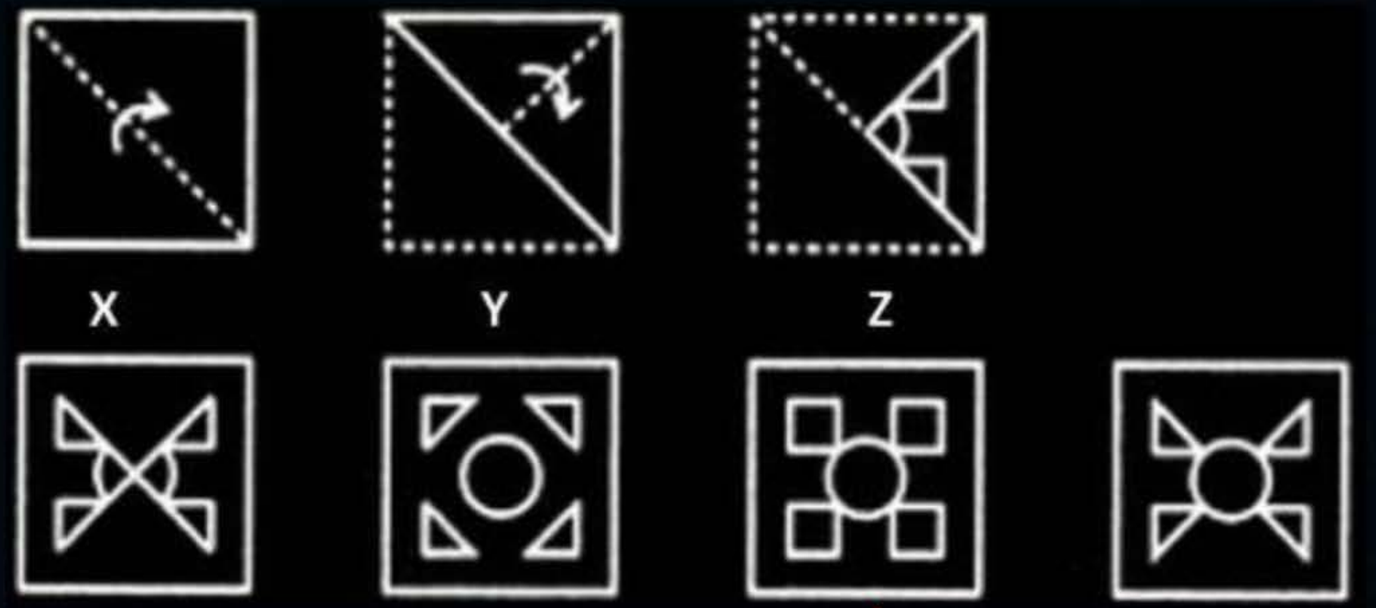
Assignment



[MCQ]

#Q. Choose a figure which would most closely resemble the unfolded form of Figure (Z).

Assignment



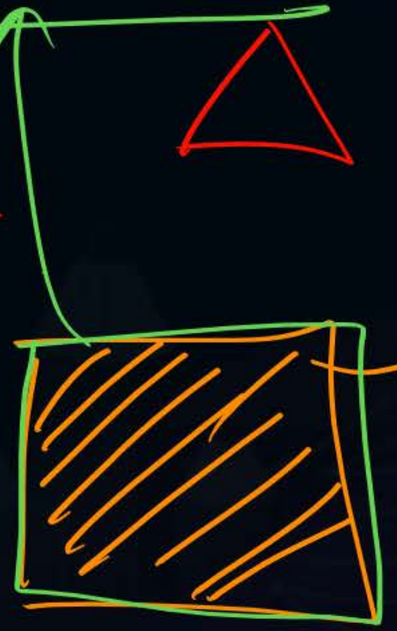
A B C D

MENSURATION: 2D

Area (A)

Perimeter (P)

2 Dimensional



3 side

Equilateral

2 side

Isosceles

Scalene

Triangles:



NORMAL

$$P = a+b+c$$

$$A = \frac{1}{2} \times b \times h$$

Heron's Formula

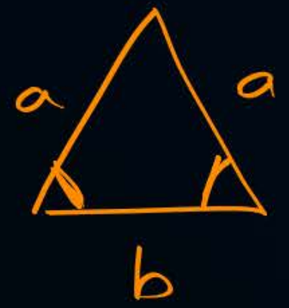
$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

where

$$s = \frac{a+b+c}{2}$$

Triangles:

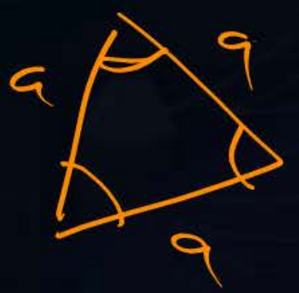
Isosceles



$$P = 2a + b$$

$$A = \frac{b}{4}\sqrt{4a^2 - b^2}$$

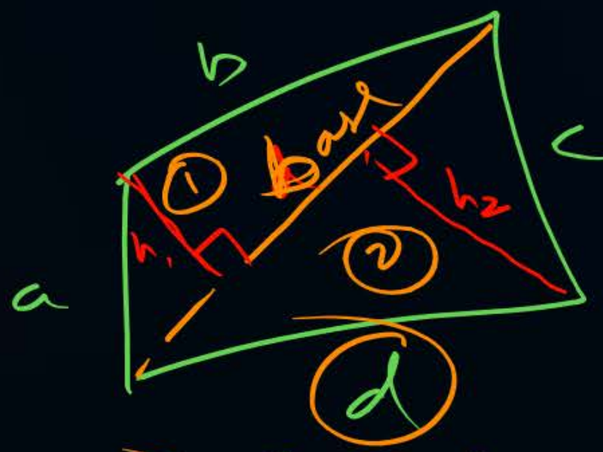
Equilateral



$$P = 3a$$

$$A = \frac{\sqrt{3}}{4} \times a^2$$

Quadilaterals:



Any

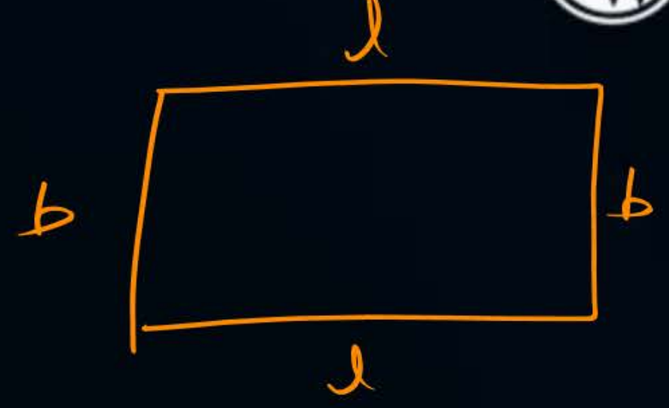
$$P = a+b+c+d \quad \text{(Adding all sides)}$$

$$A = \frac{1}{2} \times \text{base} \times h_1 + \frac{1}{2} \times \text{base} \times h_2$$

$$= \frac{1}{2} \text{base}(h_1 + h_2)$$

$$= \frac{1}{2} \times \text{base} \times (\text{Sum of Ht on the base})$$

Rectangle and Square:



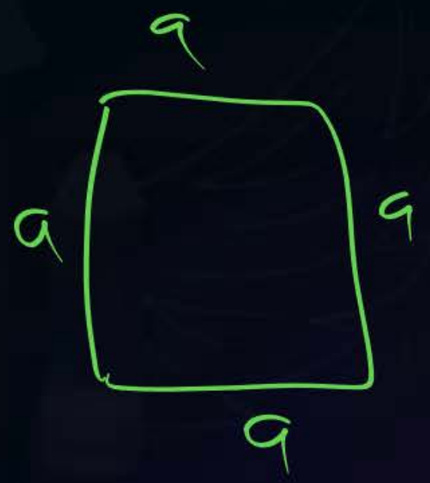
Rectangle: — $P = 2l + 2b$

$= 2(l + b)$

$A = l \times b$

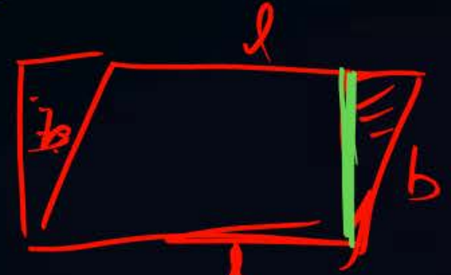
Square $P = 4a$

$A = a \times a = a^2$



Rhombus, Trapezium & Parallelogram:

Parallelogram



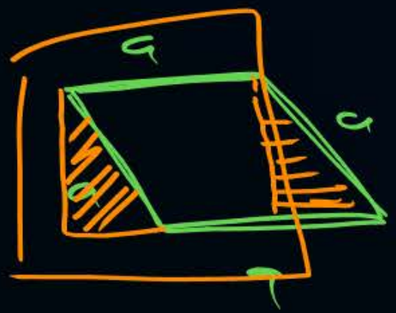
$P = 2(l+b)$

$A = l \times h$

Rhombus :-

$$P = 4a$$

$$A = a^2$$



$$\text{Area} = \frac{1}{2} \times d_1 \times d_2$$

Trapezium



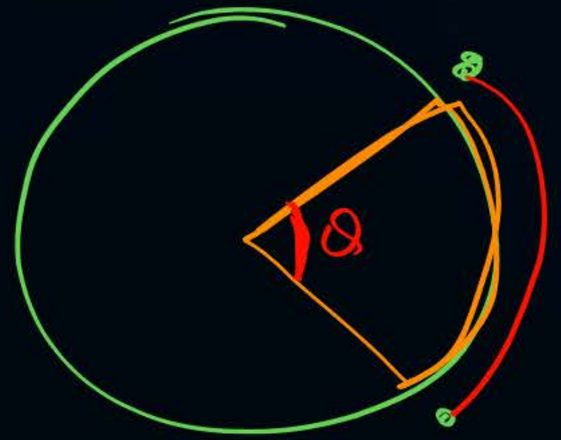
$P = a + b + c + d$

$A = \frac{1}{2}(\text{sum of parallel sides}) \times h$

Circles & Sectors:

Circumference $= 2\pi r$

Area $= \pi r^2$



Sector

$$\frac{\text{Sector}}{\text{Arc}} = \frac{\theta}{360}$$

$$\text{Length of Arc} = \frac{\theta}{360} \times 2\pi r$$

$$\text{Area of sector} = \frac{\theta}{360} \times \pi r^2$$



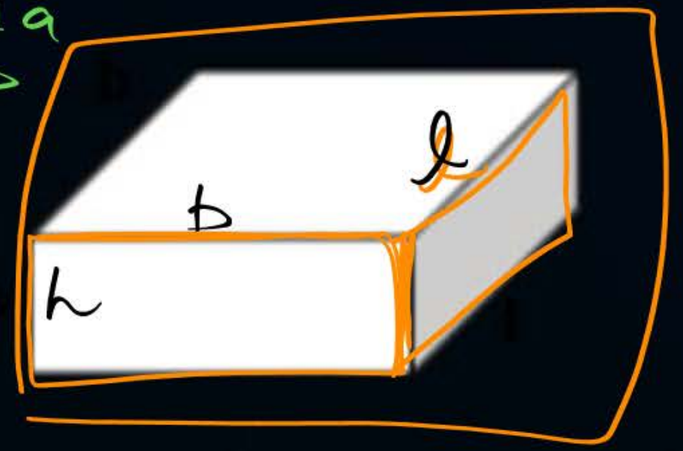
MENSURATION: 3D

3 Dimensional

Cuboid:

① Lateral Surface Area OR Curved

② Total Surface Area



Lateral surface Area

$$= 2bh + 2lh$$

$$= 2h(l+b)$$

Total surface $= 2bh + 2lh + 2lb = 2(bh + lh + lb)$

[MCQ]

#Q. What would be the area of sector of a circle whose radius is 12 cm and the length of the arc is 20 cm?

Assignment

A 60 sq. cm

B 240 sq. cm

C 120 sq. cm

D 64 sq. cm

[MCQ]

#Q. The cross-section of a canal is a trapezium in shape. If the canal is 7 m wide at the top and 9 m at the bottom and the area of cross-section is 128 sq m, find the height of the canal.

Assignment

A 32 m

B 8 m

C 16 m

D 4 m

[MCQ]

#Q. A cube of 6 cm side melted and smaller cubes of 2 cm side are manufactured. Find the number of smaller cubes so formed.

Assignment

A 12

B 27

C 24

D 8

[MCQ]

#Q. The length, breadth & height of a cuboid are in the ratio of 4 : 3 : 2 and its volume is 3000 m^3. Find its total surface area.

Assignment

A 1300 m^2

B 1500 m^2

C 1333 m^2

D 2700 m^2

[MCQ]

#Q. The dimensions of a room are 20 m x 15 m x 10 m. What is the cost of painting its four walls at the rate of ₹ 15 per 100 sq m?

Assignment

A ₹ 70

B ₹ 105

C ₹ 225

D None of these

[MCQ]

#Q. In two concentric circles, a chord length 80 cm of larger circle becomes a tangent to the smaller circle whose radius is 9 cm. The radius of the larger circle will be?

Assignment

A 13 cm

B 41 cm

C 52 cm

D 75 cm

[MCQ]

#Q. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 sq cm. The perimeter of the rectangle is?

Assignment

A 9 cm

B 18 cm

C 20 cm

D 41 cm



2 mins Summary

Topic Mensuration 2D & 3D

THANK - YOU